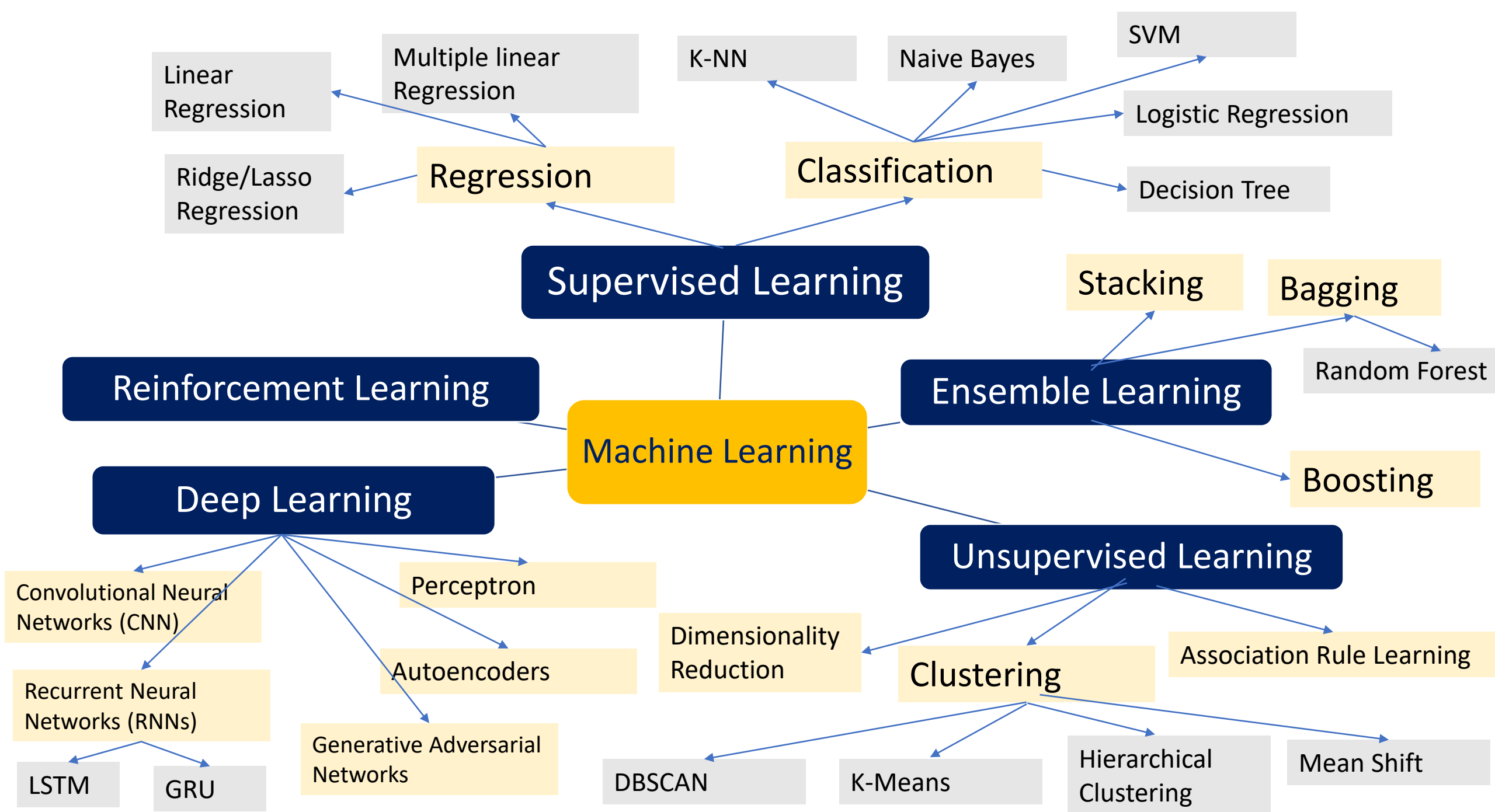


Lecture 2:

Introduction to Machine Learning

ML/AI for Engineering Applications
ENV 6932

Sara Behdad
Fall 2025



Major Machine Learning Techniques

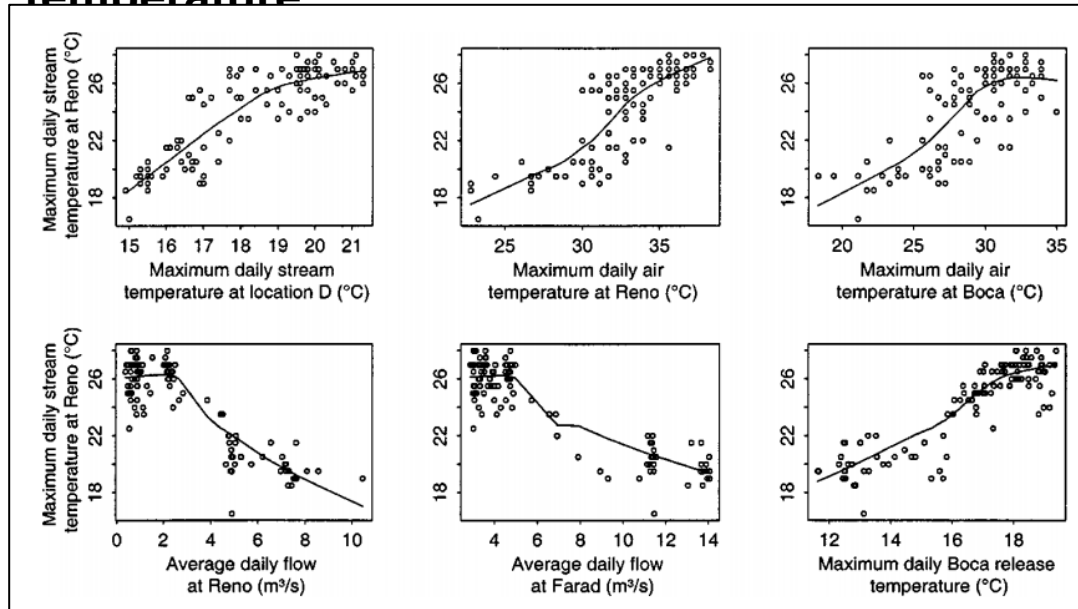
- **Regression/Estimation**
- **Classification**
- **Clustering**
- **Associations**

Major Machine Learning Techniques

- Regression/Estimation
 - Predicting continuous values
- Classification
- Clustering
- Associations

Example: Regression

Example: Predicting Truckee River **stream temperatures** at Reno by using a simple linear regression relationship based on **flow** and **air temperature**

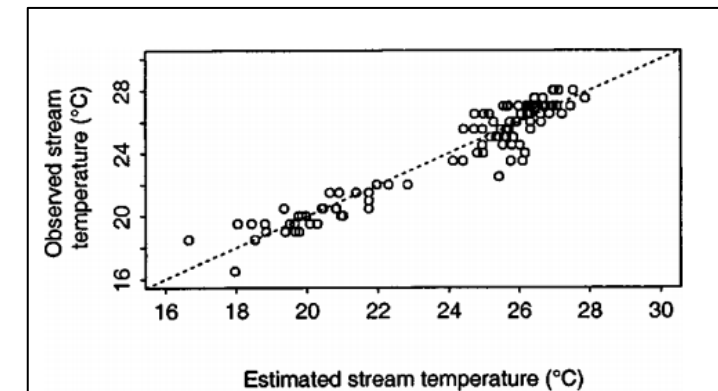
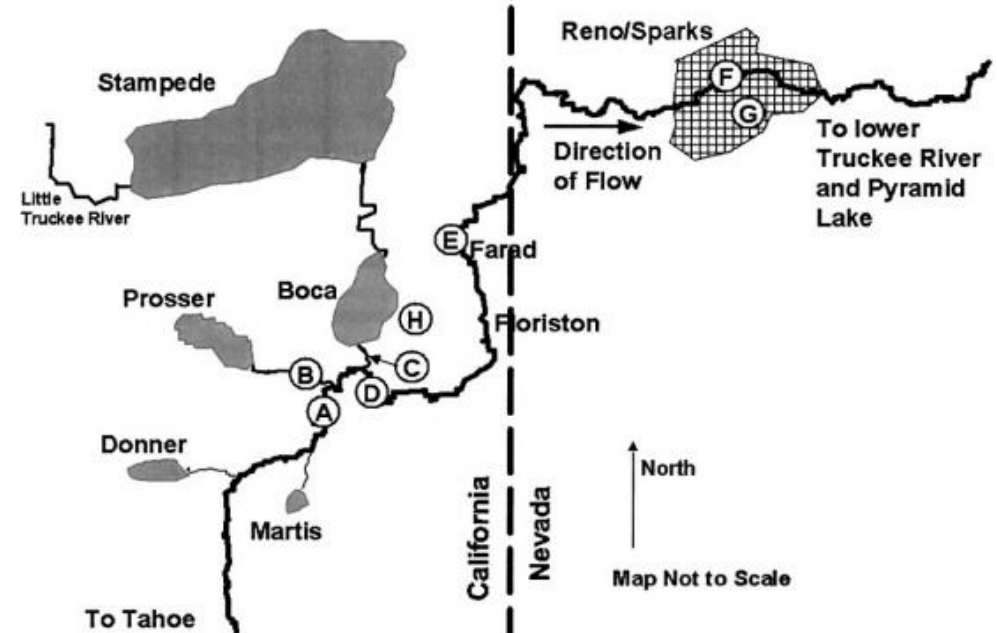


Data used as predictors

A linear regression using the predictors selected has the following equation:

$$\hat{T} = a_0 + a_1 T_{\text{Air}} + a_2 Q$$

where T_{Air} = air temperature at Reno; and Q = flow at Farad.



Major Machine Learning Techniques

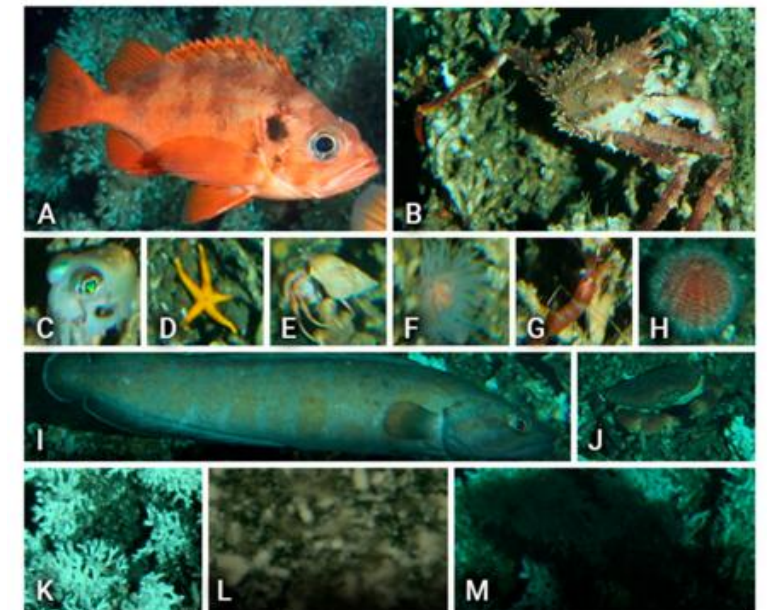
- **Regression/Estimation**
 - Predicting continuous values
- **Classification**
 - Predicting the item class/category of a case
- **Clustering**
- **Associations**

Example: Classification

- Underwater Animal Detection and Classification

An example of video-detected species used for building the training dataset for reference at automated classification.

Class (alias)	Species Name	# Specimens per Species in Dataset	Image in Figure 2
Rockfish	<i>Sebastes</i> sp.	205	(A)
King crab	<i>Lithodes maja</i>	170	(B)
Squid	<i>Sepiolidae</i>	96	(C)
Starfish	<i>Unidentified</i>	169	(D)
Hermit crab	<i>Unidentified</i>	184	(E)
Anemone	<i>Bolocera tuediae</i>	98	(F)
Shrimp	<i>Pandalus</i> sp.	154	(G)
Sea urchin	<i>Echinus esculentus</i>	138	(H)
Eel like fish	<i>Brosme brosme</i>	199	(I)
Crab	<i>Cancer pagurus</i>	102	(J)
Coral	<i>Desmophyllum pertusum</i>	142	(K)
Turbidity	-	176	(L)
Shadow	-	101	(M)



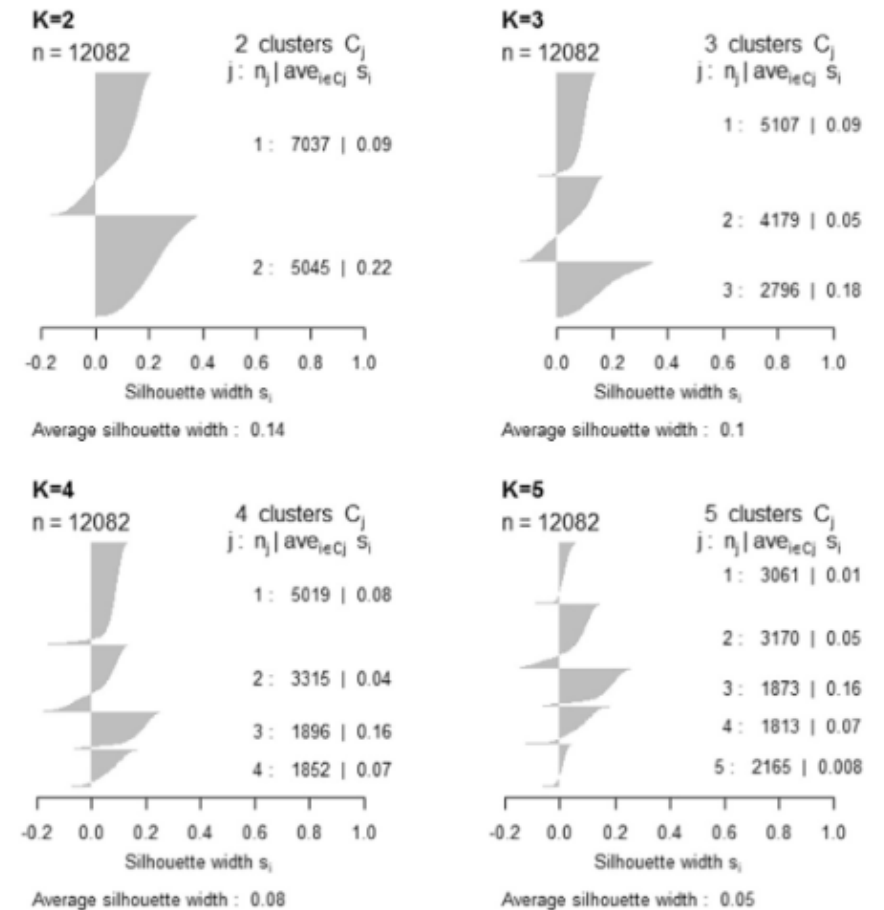
An example of video-detected species used for building the training dataset for reference at automated classification: (A) Rockfish (*Sebastes* sp.), (B) king crab (*Lithodes maja*), (C) squid (*Sepiolidae*), (D) starfish, (E) hermit crab, (F) anemone (*Bolocera tuediae*), (G) shrimp (*Pandalus* sp.), (H) sea urchin (*Echinus esculentus*), (I) eel-like fish (*Brosme brosme*), (J) crab (*Cancer pagurus*), (K) coral (*Desmophyllum pertusum*), and finally (L) turbidity, and (M) shadow.

Major Machine Learning Techniques

- **Regression/Estimation**
 - Predicting continuous values
- **Classification**
 - Predicting the item class/category of a case
- **Clustering**
 - Finding the structure of data, summarization
- **Associations**

Examples: Clustering

- Cluster households based on their energy consumption.
- **Dataset:** The 2009 Residential Energy Consumption Survey (RECS) data, collected by the US Energy Information Administration (EIA)
- How many clusters?



Cluster analysis results on the total electricity consumption data.
Silhouette analysis suggests 2 clusters.

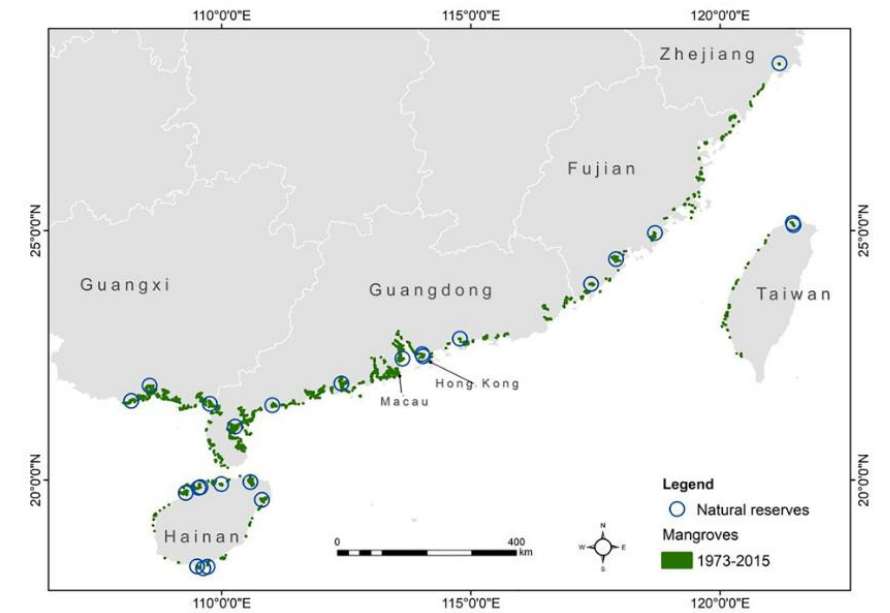
Major Machine Learning Techniques

- Regression/Estimation
 - Predicting continuous values
- Classification
 - Predicting the item class/category of a case
- Clustering
 - Finding the structure of data, summarization
- Associations
 - Associating frequent co-occurring items/events

Example: Association Rules



- **Support:** $\text{Supp}(X \Rightarrow Y) = \frac{|X \cap Y|}{N}$
- **Confidence:** $\text{Conf}(X \Rightarrow Y) = \frac{|X \cap Y|}{|X|}$
- **Lift:** $\text{Lift}(X \Rightarrow Y) = \frac{\text{Conf}(X \Rightarrow Y)}{\text{Supp}(Y)}$



The spatial distribution of total mangroves in China from 1973 to 2015. (The distribution of mangroves is shown at a coarser resolution for display purposes).

Association rules extraction for 2000–2015.

Rule ID	Policy	Urbanization	Mangroves	Support	Confidence	Lift
1	NR1	BUC0, PD1	MC2	0.019	0.765	2.478
2	NR1	BUC0	MC2	0.019	0.722	2.341
13	NR1	BUC2, PD1	MC2	0.019	0.565	1.832
14	NR1	BUC2	MC2	0.019	0.565	1.832
15	NR1	–	MC2	0.066	0.535	1.733
16	NR1	PD1	MC2	0.060	0.525	1.701
17	NR1	BUC1	MC2	0.029	0.444	1.440
18	NR1	BUC1, PD1	MC2	0.023	0.400	1.296
3	NR0	BUC1, PD0	MC0	0.011	0.667	1.181
4	NR0	BUC0, PD1	MC0	0.209	0.638	1.130
5	NR0	BUC0	MC0	0.223	0.632	1.119
8	NR0	PD1	MC0	0.489	0.592	1.049
9	NR0	–	MC0	0.517	0.590	1.045
10	NR0	BUC2, PD1	MC0	0.110	0.583	1.034
11	NR0	BUC2	MC0	0.113	0.572	1.014
22	NR0	BUC2	MC2	0.067	0.341	1.104
23	NR0	BUC2, PD1	MC2	0.061	0.326	1.056

What are Association Rules?

- Association rules uncover relationships between variables in large datasets.
- Rule format: $X \rightarrow Y$ (If X occurs, Y is likely to occur).
- Key metrics:
 - *Support: Frequency of X and Y occurring together.*
 - *Confidence: Probability of Y given X .*
 - *Lift: How much more likely Y is with X than without.*

Example: Urbanization and Mangrove

Hypothetical Dataset (100 coastal grid cells in China)

- 35 cells: High urbanization + mangrove loss
 - 15 cells: High urbanization + mangrove intact
 - 20 cells: Low urbanization + mangrove loss
 - 30 cells: Low urbanization + mangrove intact
-
- **Rule to Test:** High Urbanization $\Rightarrow$ Mangrove
 - **Support:**

$$\text{Support} = \frac{\text{Cells with High Urbanization AND Mangrove Loss}}{\text{Total cells}} = \frac{35}{100} = 0.35$$

Example: Urbanization and Mangrove

Hypothetical Dataset (100 coastal grid cells in China)

- 35 cells: High urbanization + mangrove loss
- 15 cells: High urbanization + mangrove intact
- 20 cells: Low urbanization + mangrove loss
- 30 cells: Low urbanization + mangrove intact

- **Confidence:**

$$\text{Confidence} = \frac{\text{Cells with High Urbanization AND Mangrove Loss}}{\text{Cells with High Urbanization}} = \frac{35}{(35 + 15)} = \frac{35}{50} = 0.70$$

- **Lift:**

$$\text{Lift} = \frac{\text{Confidence}}{\text{Support(Mangrove Loss)}}$$

$$\text{Support(Mangrove Loss)} = (35+20)/100 = 0.55$$

$$\text{Lift} = 0.70 \div 0.55 \approx \mathbf{1.27}$$

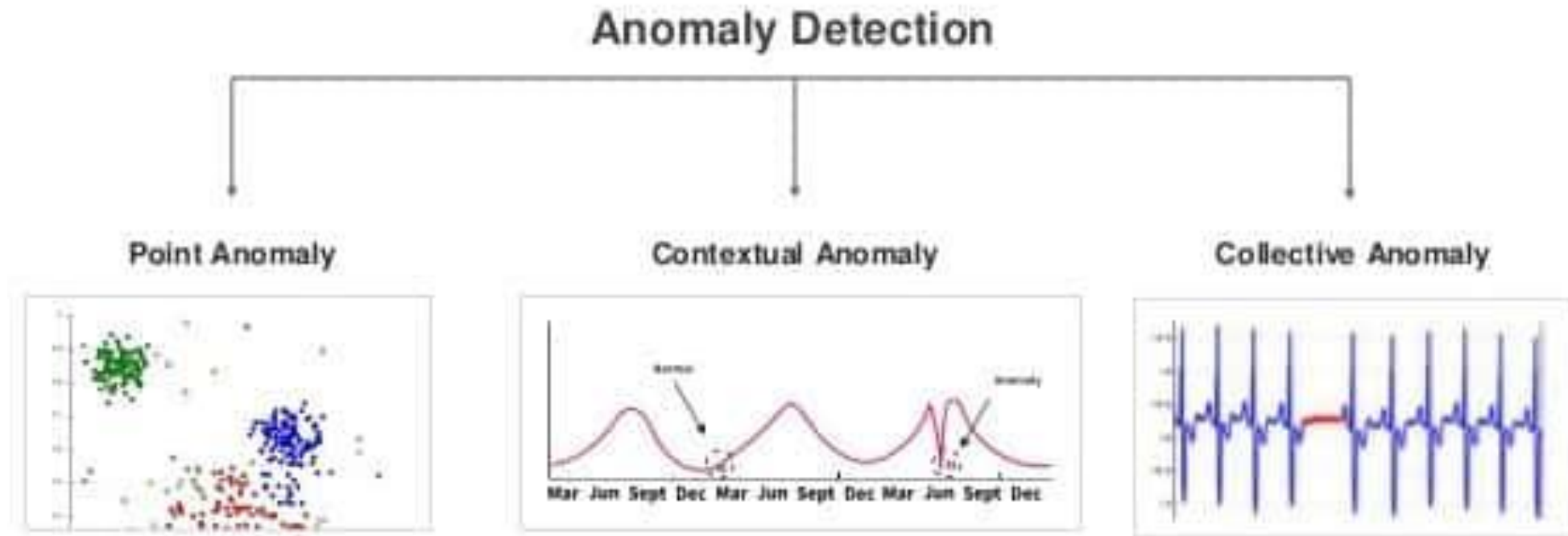
Major Machine Learning Techniques

- **Anomaly Detection**
- **Sequence Mining**
- **Dimension Reduction**
- **Recommendation Systems**

Major Machine Learning Techniques

- **Anomaly Detection**
 - Discovering abnormal or unusual cases
- **Sequence Mining**
- **Dimension Reduction**
- **Recommendation Systems**

Example: Anomaly detection



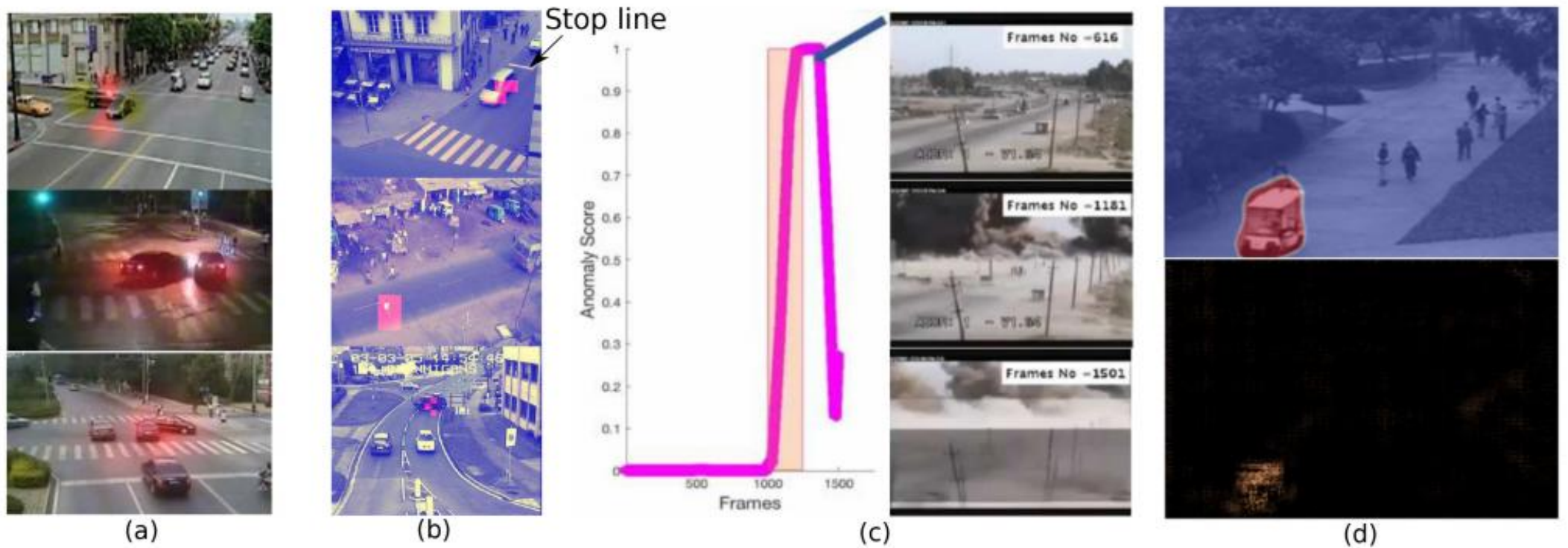
Types of anomalies: a) point anomaly; b) contextual anomaly; and c) collective anomaly

Types of Anomalies in Machine Learning

- a) **Point Anomaly (Global Anomaly)**
 - A single data instance is anomalous compared to the rest.
 - *Example:* A transaction of \$10,000 when most are <\$200.
- b) **Contextual Anomaly (Conditional Anomaly)**
 - A data instance anomalous in a specific context but normal in another.
 - *Example:* 20°C temperature is normal in summer but anomalous in winter.
- c) **Collective Anomaly**
 - A group of data instances is anomalous together, even if individual points are not.
 - *Example:* A burst of small but rapid credit card charges indicating fraud.

Example: Anomaly detection

- Road Traffic



Visual snapshots of some of the state-of-the-art anomaly detection techniques

Major Machine Learning Techniques

- **Anomaly Detection**
 - Discovering abnormal or unusual cases
- **Sequence Mining**
 - Predicting next event, click-stream (Markov models)
- **Dimension Reduction**
- **Recommendation Systems**

Example: Website Design

- **An important input to the website design:** the analysis of how a Web site is being used.
- **Web usage mining:** finding the common traversal paths through a web site
- **Three different ways of identifying transactions:**
 - Reference Length: time spent when visiting a page
 - Maximal Forward Reference: set of pages in the path from the first page in a user session up to the page before a backward reference is made
 - Time Window.

Example: Website Design

- Olympics Web site examples:
 - 9.81% of the site visitors accessed the Atlanta home page followed by the Sneakpeek main page.
 - 0.42% of the site visitors accessed the Sports main page followed by the Schedules main page

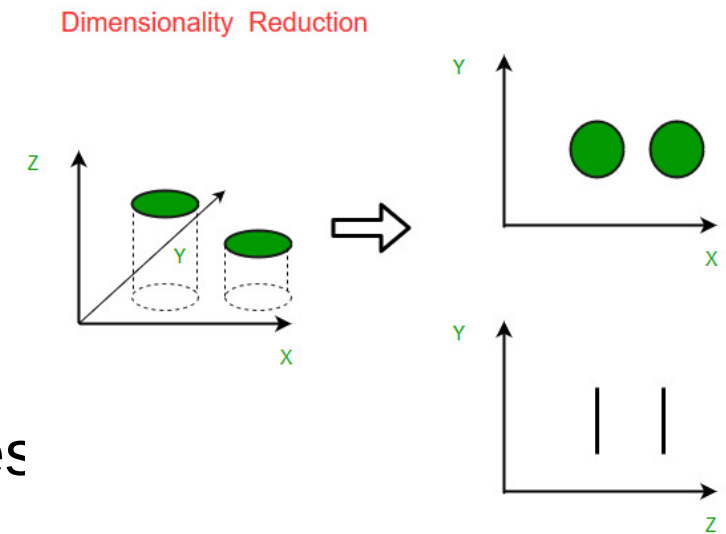


Major Machine Learning Techniques

- **Anomaly Detection**
 - Discovering abnormal or unusual cases
- **Sequence Mining**
 - Predicting next event, click-stream (Markov models)
- **Dimension Reduction**
 - Reducing the size of data (PCA: Principal Component Analysis)
- **Recommendation Systems**

Example: Dimension Reduction

- **Dimensionality reduction:** techniques for reducing the number of input variables in training data.
- **Benefits:** better human interpretations and low computational cost
- **Applications:**
 - noise reduction
 - data visualization
 - cluster analysis
 - an intermediate step to facilitate other analyses



Dimensionality Reduction Techniques

(1) Feature Selection

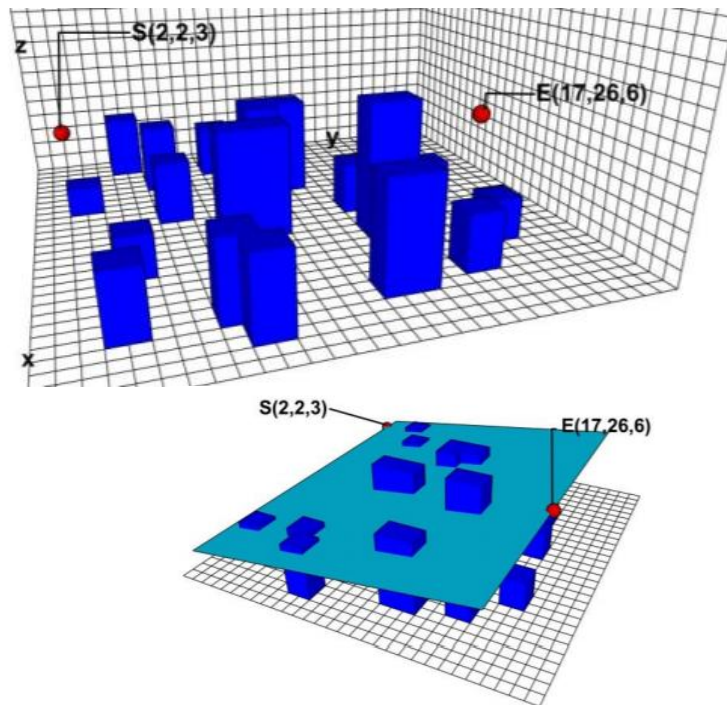
- Choose a subset of original features.
- **Methods:** correlation, mutual information, LASSO regression.
- *Example:* Remove redundant variables.

(2) Feature Extraction

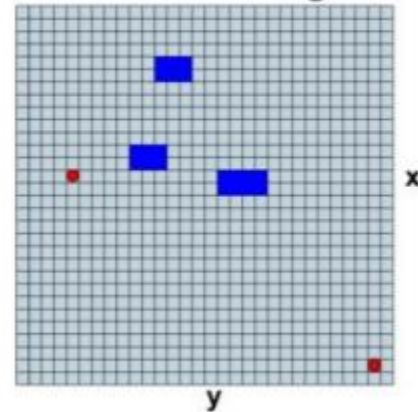
- Create new features from original data.
- **Methods:** PCA, autoencoders.
 - PCA - linear projection maximizing variance.
 - Autoencoders - neural networks that learn compressed latent features.
- *Example:* Reduce image pixels to fewer components.

Example: Dimension Reduction

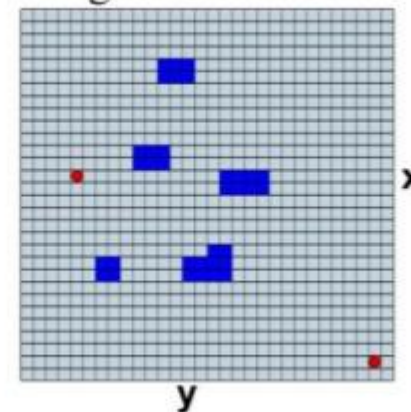
- Path planning of unmanned aerial vehicle (UAV) using an environment modeling method with height dimension reduction
- Planning algorithm could avoid unnecessary calculations



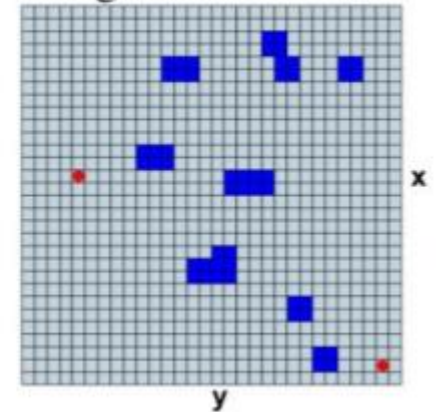
The Same Height



Higher Start Point



Higher End Point

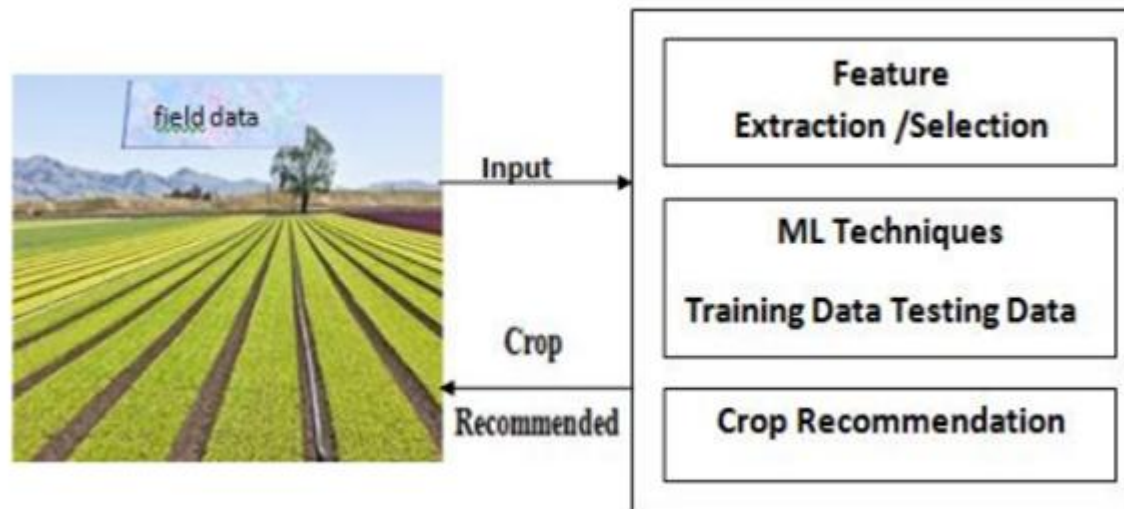


Major Machine Learning Techniques

- **Anomaly Detection**
 - Discovering abnormal or unusual cases
- **Sequence Mining**
 - Predicting next event, click-stream (Markov models)
- **Dimension Reduction**
 - Reducing the size of data (PCA)
- **Recommendation Systems**
 - Recommending items

Example: Recommendation Systems

- **Intelligent crop recommendation system**
- picking an appropriate crop for their farming land based on the essential variables (geological and climatic attributes).

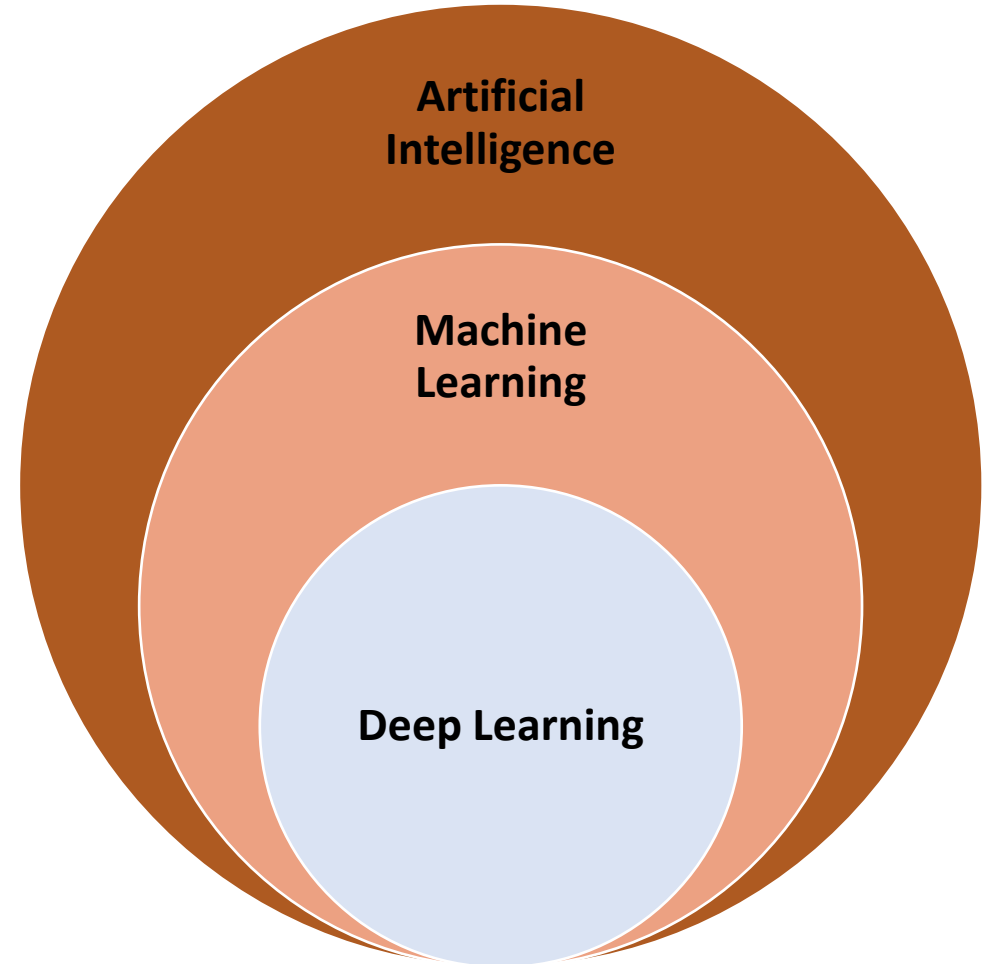


Crop	Accuracy (%)
Onion	94.78
<u>Ragi</u>	93.60
Rice	92.04
Potato	87.69
Gram	85.43

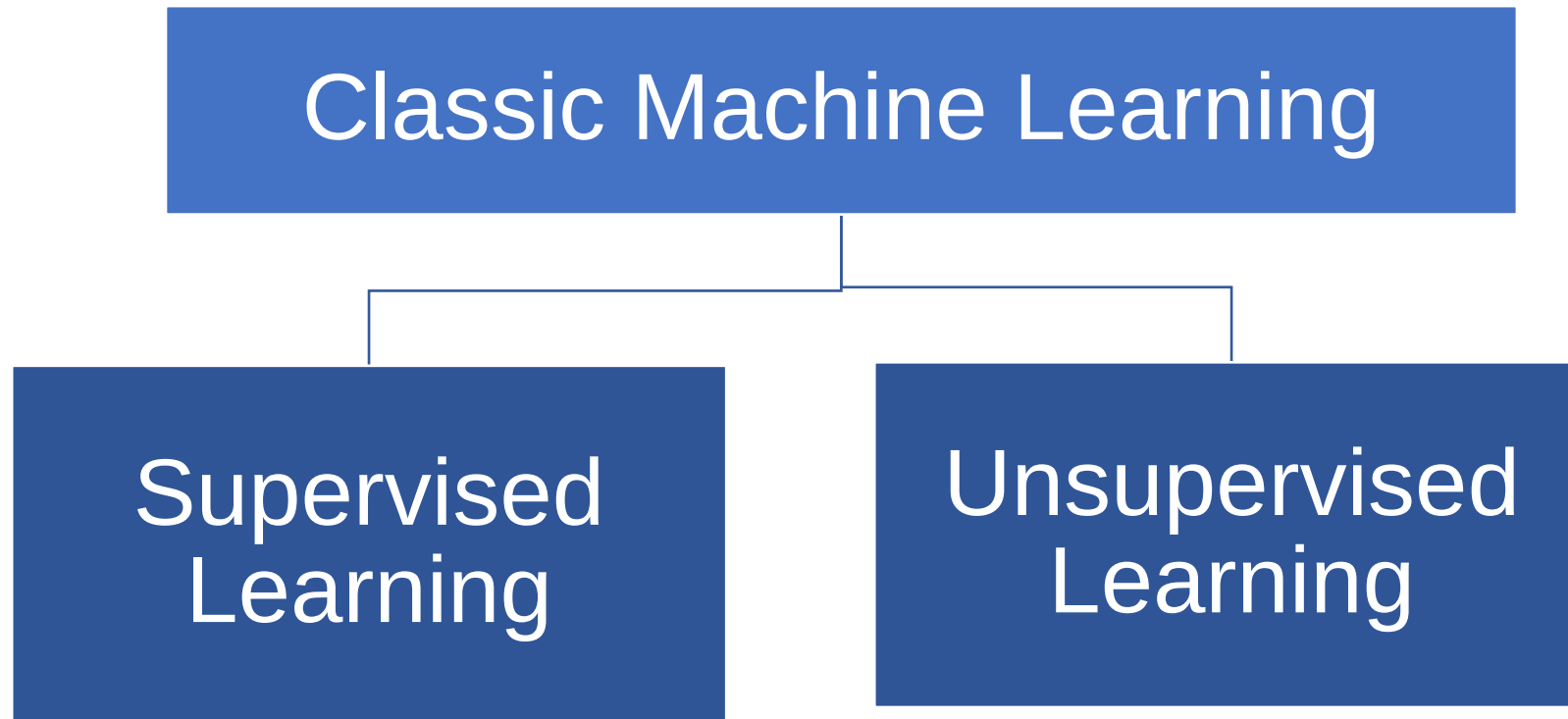
<https://www.prolim.com/crop-recommendation-system-using-machine-learning-for-digital-farming/>

Difference between Artificial Intelligence, Machine Learning and Deep Learning

- **AI components**
 - Computer vision
 - Language processing
 - Creativity
 - etc
- **Machine Learning**
 - Classification
 - Clustering
 - Neural network
 - etc
- **Revolution in ML**
- Deep Learning



Two Major Groups of Machine Learning Techniques



Supervised Learning

- How do we supervise machines and algorithms?
- We teach the model
- How?
- We teach the model by training it with some data from a labeled dataset.

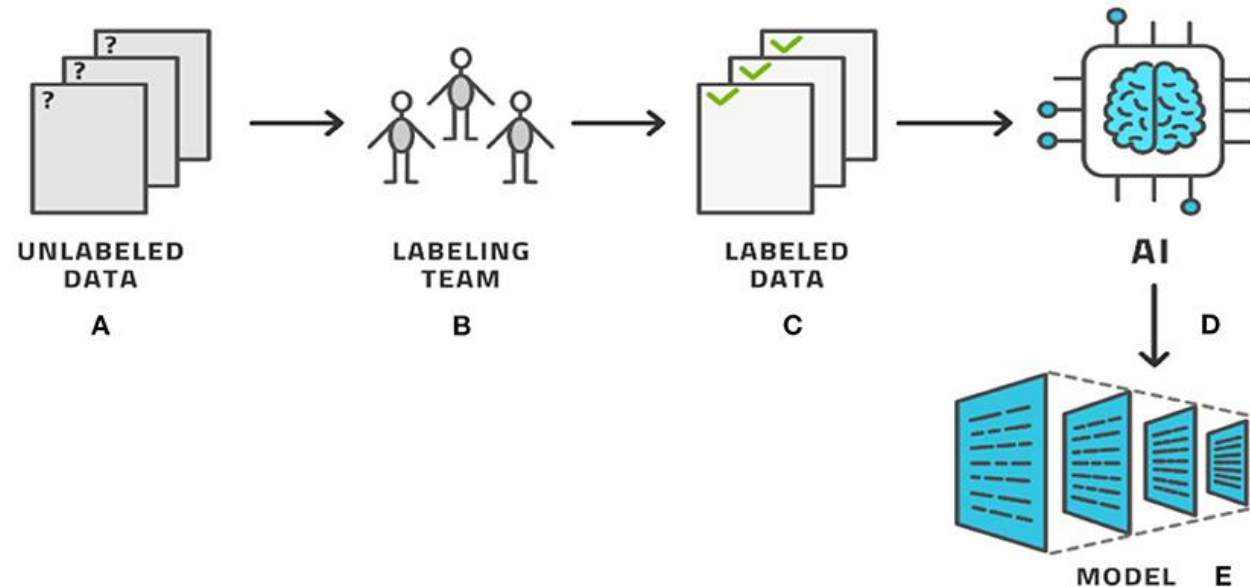
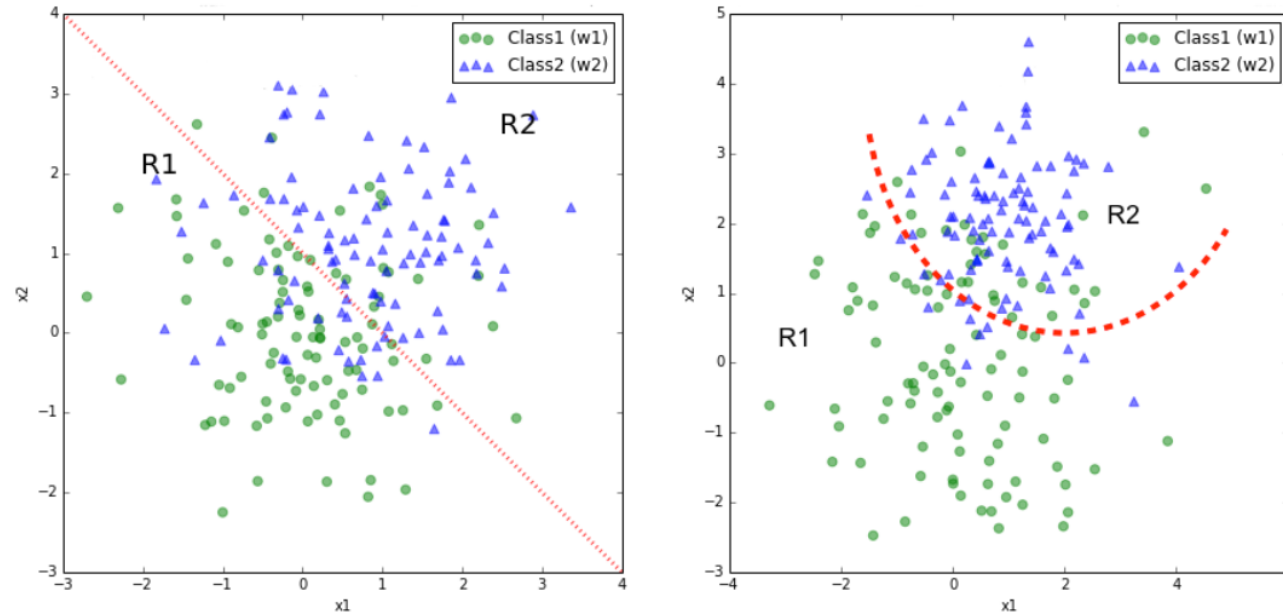


Image Reference: Bangert, Patrick, et al. "Active learning performance in labeling radiology images is 90% effective." *Frontiers in Radiology* 1 (2021): 748968.

Supervised Learning



an exemplary classification task for samples with two random variables

we “teach the model,”
then with that
knowledge, it can
predict unknown or
future instances

Supervised Learning

- The famous “Iris” flower dataset created and used by R. A. Fisher, and it is freely [available](#) at the UCI machine learning repository.
- It serves as a perfect example of a supervised classification task
- Three flower species/labels: Setosa, Virginica, and Versicolor.
- And every of the 150 instances (individual flowers) consists of four features:
 - Sepal width
 - Sepal length
 - Petal width
 - Petal height

Iris Data Set

Download: [Data Folder](#), [Data Set Description](#)

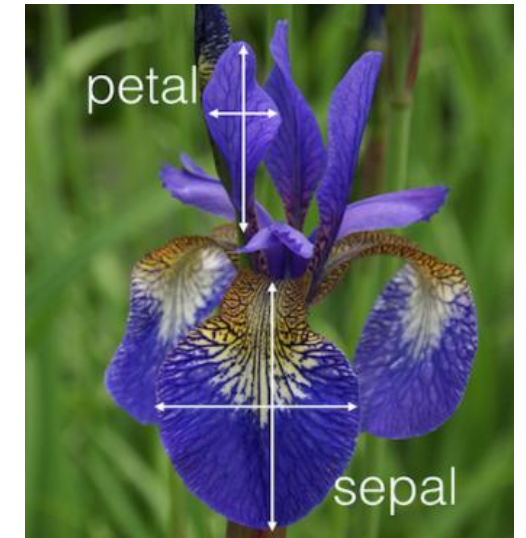
Abstract: Famous database; from Fisher, 1936



Data Set Characteristics:	Multivariate	Number of Instances:	150	Area:	Life
Attribute Characteristics:	Real	Number of Attributes:	4	Date Donated	1988-07-01
Associated Tasks:	Classification	Missing Values?	No	Number of Web Hits:	4154715

Teaching the Model with Labeled Data

	Sepal length	Sepal width	Petal length	Petal width	Class
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
⋮	⋮	⋮	⋮	⋮	⋮
150	5.9	3.0	5.1	1.8	virginica



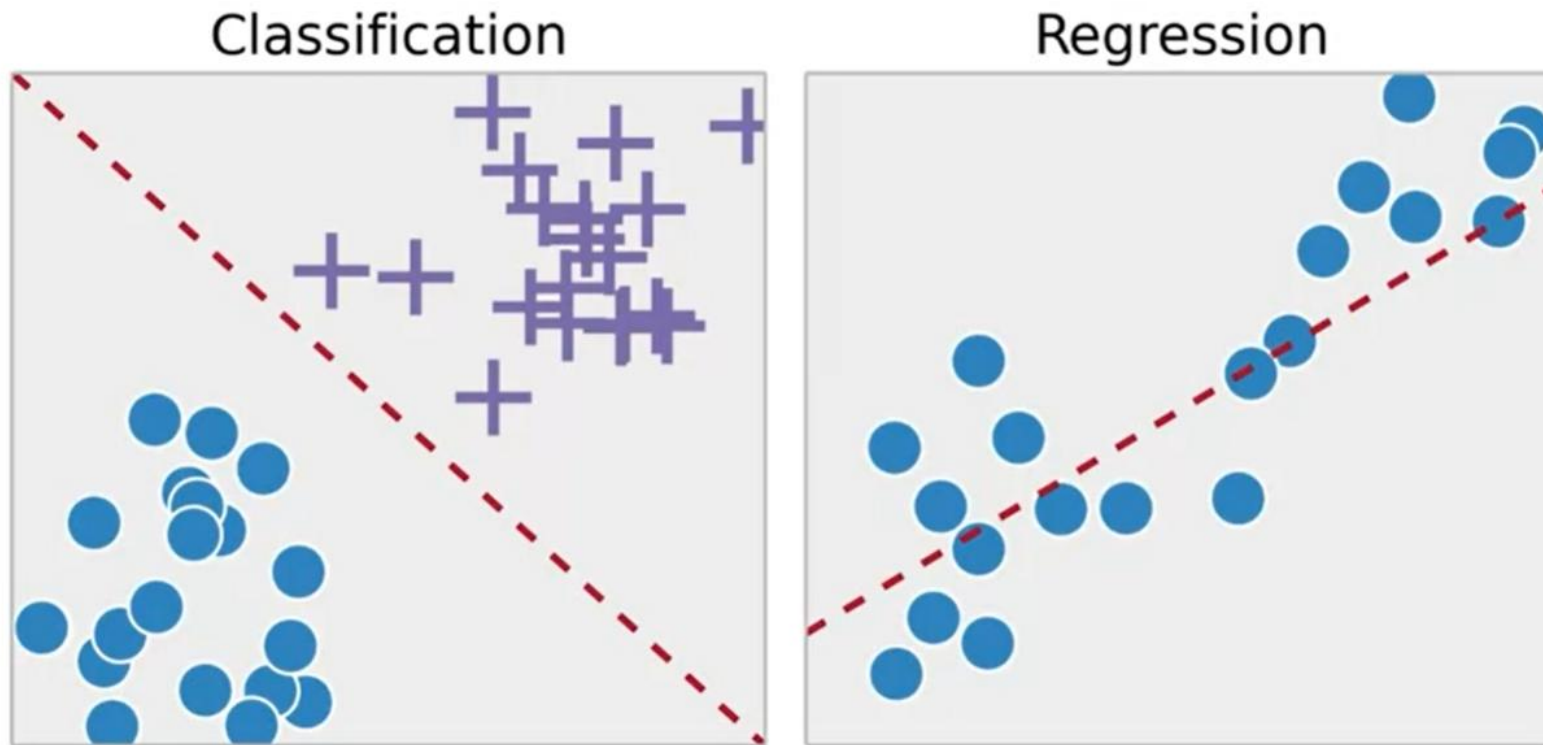
Bloice, M. D., & Holzinger, A. (2016). A tutorial on machine learning and data science tools with python. *Machine Learning for Health Informatics*, 435-480.

Two Types of Data: Numerical vs. Categorical

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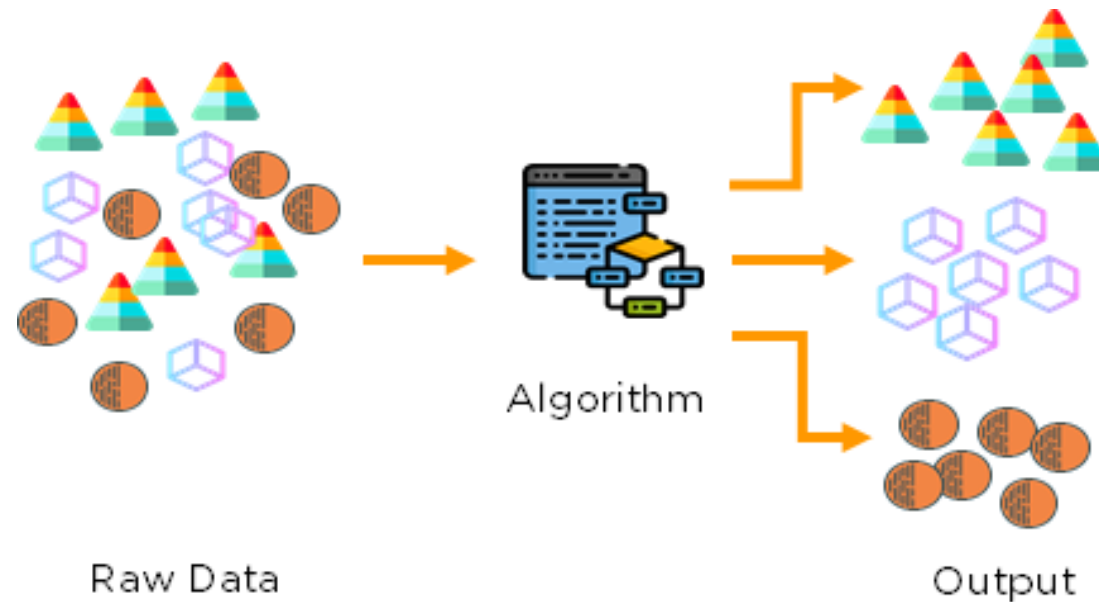
Bloice, M. D., & Holzinger, A. (2016). A tutorial on machine learning and data science tools with python. *Machine Learning for Health Informatics*, 435-480.

Two Examples of Supervised Learning



Unsupervised Learning

- We do not supervise the model.
- The model works on its own to discover the info that may not be visible to human.
- The unsupervised algorithms **trains on unlabeled dataset.**



What is Unsupervised Learning?

	Sepal length	Sepal width	Petal length	Petal width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
⋮	⋮	⋮	⋮	⋮
150	5.9	3.0	5.1	1.8

The model works on its own to discover information



ALL OF THIS DATA IS UNLABELED

What is Unsupervised Learning?

- **Unsupervised learning techniques**

- Dimension reduction,
- Density estimation,
- Market basket analysis,
- Clustering

	Sepal length	Sepal width	Petal length	Petal width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
⋮	⋮	⋮	⋮	⋮
150	5.9	3.0	5.1	1.8

The model works on its own to discover information

ALL OF THIS DATA IS UNLABELED



Supervised vs. Unsupervised Learning

Supervised Learning

- **Classification:** Classifies labeled data
- **Regression:** Predicts trends using previous labeled data
- Has more evaluation methods than unsupervised learning
- Controlled environment

Unsupervised Learning

- **Clustering:** Finds patterns and groupings from unlabeled data
- Has fewer evaluation methods than supervised learning
- Less controlled environment

Difference Between Supervised and Unsupervised Learning

- The difference is the **problem statement**.
- **Supervised learning:** we have two types of variables: (1) dependent variable or target variable (y), and (2) independent variables or explanatory variables (x)
- **Unsupervised models:** there is no split between target and independent variables. We aim to group data points by analyzing their similarity.